

Method of indicator variables

1. Imagine the chance experiment of drawing a hand of five cards from a standard deck. Let N be the random variable that counts the number of hearts in your hand.

a. For $i=1,2,\dots,5$, let A_i be the event that card i in your hand is a heart. Calculate $P(A_i)$.

$$P(A_i) = \frac{1}{5}$$

b. Let X_i be the indicator variable for the event A_i . This means that X_i has the value 1 for any outcome where A_i occurs and that X_i has the value 0 for any outcome where A_i does not occur. Calculate $E[X_i]$.

$$E[X_i] = \frac{1}{5}$$

c. We can write N as $N = X_1 + X_2 + X_3 + X_4 + X_5$: the sum on the right-hand side counts the number of events A_i that occur, which is exactly the value of N . Use this formula for N to calculate $E[N]$.

2. Here is another way to calculate $E[N]$

a. Label the hearts in the deck with numbers 1, 2, 3, ..., 13. (It does not matter how you do this.) For $i=1,2,3,\dots,13$, let B_i be the event that heart card i is in your hand. Calculate $P(B_i)$.

$$P(B_i) = \frac{1}{13}$$

b. Let Y_i be the indicator variable for the event B_i . This means that Y_i has the value 1 for any outcome where B_i occurs and that Y_i has the value 0 for any outcome where B_i does not occur. Calculate $E[Y_i]$.

$$E[Y_i] = \frac{1}{13}$$

c. We can write $N = Y_1 + Y_2 + \dots + Y_{13}$: the sum on the right-hand side counts the number of events B_i that occur, which is exactly the value of N . Use this formula for N to calculate $E[N]$.